

Exact Labelled Layer-Order Classification for Period-2 Mountain–Valley Assignments on $2 \times 2k$ Maps

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Abstract

We study exact labelled layer orders for prescribed period-2 mountain–valley assignments on ordinary orthogonal $2 \times 2k$ maps. A state records a strict total order of the labelled faces; fold sequences and symmetry classes are not counted. For $k \geq 2$, the standard local flat-foldability condition partitions the 64 formal six-bit words into 48 locally obstructed and 16 locally admissible words. The admissible words split into an eight-word pleat regime with singleton state sets and two four-word constant-row regimes. For canonical $MMMMVV$, upper/lower pivot compatibility is exactly $a = 1$, $a = m$, or $a = b$, where $m = 2k - 1$; for canonical $MVMMMM$, it is exactly $B_R(a) = B_R(b)$, with $B_R(1) = 0$ and $B_R(p) = \lfloor p/2 \rfloor$ for $p \geq 2$. Exact labelled symmetry transports these canonical fibres to every admissible period word, and each compatible ordered pivot pair determines one labelled state. The resulting four state-set regimes have cardinalities 0, 1, $6k - 5$, and $4k - 3$ as a downstream corollary. The boundary $k = 1$ is treated separately: after projection to the four physical crease bits, the state set is empty or a singleton according to δ_0 . Bounded exact computation independently validates finite instances and the supplementary tables, but the all- k results follow from the symbolic arguments.

1 Introduction

Once a rectangular map is placed flat, feasibility still depends on the vertical ordering of overlapping labelled faces. For prescribed mountain–valley assignments, local crease directions induce precedence relations, while global nonintersection imposes additional nesting constraints on the resulting order [1, 2]. This makes the layer-order set itself a natural object: existence asks whether the set is empty, whereas enumeration records only its cardinality.

We consider ordinary orthogonal $2 \times 2k$ maps whose mountain–valley labels repeat with period two in the horizontal, upper-vertical, and lower-vertical crease families. The formal assignment is a six-bit word $P = (h_0, h_1, u_0, u_1, l_0, l_1)$, but these six bits are periodic labels rather than six physical creases. Our primary object is the exact set $F(P, k)$ of globally valid strict total orders of all labelled faces, with no quotient by symmetry and no counting of fold sequences.

Period two is restrictive enough that, in the constant-row branches, each physical row reduces to a one-parameter pivot normal form, while the interaction between the two row orders remains global and produces two distinct compatibility mechanisms. This balance is what makes an explicit state-set description possible without reducing the problem to a purely local count.

For $k \geq 2$, the classification is organized by the sign invariants $\delta_i = h_0 h_1 u_i l_i$, $\tau = u_0 u_1$, and $\rho = h_0 h_1$. A standard local condition eliminates 48 of the 64 formal words, and exact labelled symmetries reduce the remaining global analysis to pleat states and two canonical constant-row families. The $MMMMVV$ family is governed by a boundary-or-diagonal pivot criterion, whereas

the *MVMMMM* family is governed by equality of a two-site block coordinate. These mechanisms assemble into an exact four-regime state-set theorem, with the formulas 0 , 1 , $6k - 5$, and $4k - 3$ appearing only afterward as cardinality consequences.

We use established final-state validity and order-extension frameworks, together with the known uniform-sign strip family, as starting points for the analysis below. Proposition 6.1 records the row orders in explicit seam-indexed form and gives a self-contained derivation in the notation needed for the two-row problem; no priority claim is made for the underlying one-dimensional classification.

The family-specific results proved here begin with the interaction of the two row states. We give exact labelled transports among the period words, explicit compatibility criteria for the two constant-row canonical families, and a multiplicity-one correspondence between compatible ordered pivot pairs and labelled states after the remaining two-dimensional nonintersection conditions are verified. The all-word state-set classification is then assembled from those fibres, and the count formulas are downstream cardinality consequences. The width-two case $k = 1$ is handled separately because two formal period bits have no physical crease carriers there.

The proof follows the same structural order as the statement: local reduction, exact symmetries, the pleat singleton mechanism, row-local pivot normal forms, the two canonical constant-row compatibility theorems, and final all-word synthesis. Technical interval and uniqueness arguments are retained in the appendices so that the main text can display the mechanisms without replacing proof obligations by intuition. Finite exact enumeration is used as an independent validation tool, not as a premise for any all- k statement.

2 Related work

2.1 Layer-order validity for rectangular maps

Nishat gives a validity characterization for labelled rectangular-map orders that combines the directed mountain–valley precedence network with twin-butterfly stacking or nesting conditions, together with an exact-order validation procedure [1]. Nishat and Whitesides provide a shorter $2 \times n$ publication-lineage treatment of the same map-folding setting [2]. We use these ingredients as background semantics, including the degree-four local flat-foldability condition whose period-two specialization yields the two δ_i equations [1, 2].

2.2 Direct $2 \times n$ and all-state frameworks

Morgan’s 2012 thesis gives a polynomial-time algorithm for deciding flat foldability of general assigned $2 \times n$ maps and, in its elementary cases, already supplies existence for the two locally admissible structural branches that arise after the period-two local reduction [3, Section 2.2.2]. Thus nonemptiness of the locally admissible period-two assignments is not claimed here as new; the issue studied below is their exact labelled state sets. Jia and Mitani formulate fixed-MV $2 \times n$ folding as an order-extension problem over a $1 \times 2n$ strip representation with map-specific nonintersection filtering [4]. Their fixed-assignment procedure is an existence/decision framework; their broader enumeration procedure enumerates MV patterns rather than the multiplicity of every labelled layer order of one fixed pattern. Jia, Mitani, and Uehara also study the related $2 \times n$ box-pleated geometry without a prescribed mountain–valley assignment [5]. These works provide direct algorithmic context but do not state the period-two fibre criteria proved below.

Akitaya, Demaine, and Ku give a broader facewise layer-order constraint formulation, decompose the residual constraint system into independent components, and preserve multiplicity through Cartesian-product assembly of component solutions [6]. Accordingly, general exact-state enumeration

and multiplicity-preserving decomposition are treated here as established framework-level ideas rather than as contributions of the period-two specialization.

2.3 Fixed-MV strips and pleat context

One-dimensional fixed-MV strip folding has an established counting and structural literature. Asano et al. prove that a unit-spaced strip has a unique folded state if and only if its MV pattern is a pleat [7]. For the uniform-sign family, Hull et al. prove $c(M^q) = c(V^q) = q$, describe the corresponding two-spiral states, and use the adjacent pair of extreme faces as the structural parameter [8, Theorem 2.1 and the proof of Theorem 3.1]. With $q = 2k - 1$, this is the one-dimensional family underlying Proposition 6.1; the formulas $R^\pm(k, p)$ below are an explicit seam-indexed reformulation chosen for the subsequent two-row analysis, not a new one-dimensional classification. Hoshido et al. provide later fixed-MV strip context [9]. Uehara’s 2011 chapter and Umesato et al. 2013 remain part of that historical literature [10, 11]; we do not use their unavailable full bodies as sole proposition-level authority. We therefore make no priority claim for either the underlying uniform-sign row family or the pleat mechanism.

2.4 Algebraic layer-order work

The fixed-row horizontal reconstruction used in the *MVMMMM* branch is an instance of the classical single-stack problem; in permutation language, the generic three-label obstruction is the standard 231 obstruction to one-stack sortability [12]. The work specific to the present family is the evaluation of that stack condition on the pivot-indexed row words, together with the proof that a surviving horizontal merge satisfies the remaining two-dimensional map constraints.

Ku, Terao, and Terao have a recent algebraic treatment of layer-order constraints in the same broad methodological lineage [13]. The full primary chapter body was not accessible to us for proposition-level comparison, so we do not infer which of the explicit period-two statements may follow from that work and make no priority claim relative to it. The distinction maintained throughout this paper is therefore between established general validity, strip, stack, and solver frameworks and the explicit symbolic specialization proved here, not between possible and impossible prior approaches.

3 Model, semantics, and period-2 notation

Figure 1 summarizes the physical face and crease notation used below; the definitions remain textual and do not rely on the schematic.

Definition 3.1 (Exact labelled layer-order states). Fix $k \geq 2$ and put $n = 2k$. The upper faces are $A_1, \dots, A_n$ and the lower faces are $B_1, \dots, B_n$. The physical creases are

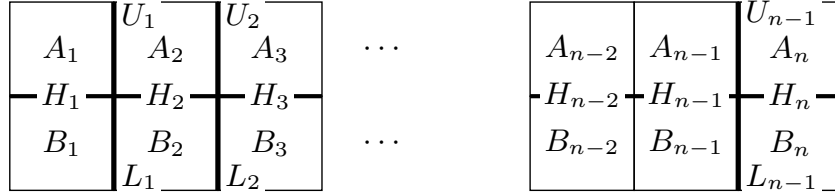
- $H_c = (A_c, B_c)$ for $1 \leq c \leq n$;
- $U_i = (A_i, A_{i+1})$ for $1 \leq i \leq n - 1$;
- $L_i = (B_i, B_{i+1})$ for $1 \leq i \leq n - 1$.

A period-2 mountain–valley assignment is encoded by

$$P = (h_0, h_1, u_0, u_1, l_0, l_1) \in \{M, V\}^6.$$

The crease with physical index j uses period slot $(j - 1) \bmod 2$: H_c receives $h_{(c-1) \bmod 2}$, and U_i, L_i receive $u_{(i-1) \bmod 2}, l_{(i-1) \bmod 2}$.

$2 \times n$ **physical map**, $n = 2k$



period slot: physical index j uses $(j - 1) \bmod 2$

targets: $H \mapsto S$; odd vertical seams $U/L \mapsto E$; even vertical seams $U/L \mapsto W$

Figure 1: Schematic of the physical $2 \times n$ map ($n = 2k$). Faces remain labelled. H_c is the horizontal crease between A_c and B_c ; U_i and L_i are the physical shared seams between adjacent upper and lower faces. Physical index j uses period slot $(j - 1) \bmod 2$. The six formal period bits specify repeating labels, not six physical creases.

The validity semantics below are the specialization of the labelled-order framework for rectangular map folding [1, 2]. A physical flat-folded state of the prescribed map is represented by a strict labelled total order of all $4k$ faces. Two logically distinct requirements apply. First, the prescribed crease assignment itself must satisfy ordinary local flat-foldability at every present interior vertex; this is a prerequisite on the assignment P , independent of any proposed total order. Throughout, $x \prec y$ means that face x lies above face y ; displayed total orders are written from top to bottom. We consider final zero-thickness layer orders in this rectangular-map model, not the existence of a prescribed continuous folding motion. Second, a candidate total order must satisfy the following three *global order-validity* conditions.

1. **Directed M/V precedence.** Identify A_c with row 0 and B_c with row 1. A face (r, c) is light-up exactly when $(r + c - 1) \bmod 2 = 0$. Across an M crease, the light-up wing precedes the dark-up wing; across a V crease, the dark-up wing precedes the light-up wing.
2. **Folded target classes.** Every H_c has target class S . Each U_i and L_i has target class E for odd i and W for even i . The symbols S, E, W identify target classes in the folded geometry: only creases assigned to the same target class can enter the same disjoint-wing Butterfly test in the next clause. Thus S collects horizontal creases, while E and W distinguish the two vertical-seam parities.
3. **Butterfly/noncrossing validity.** For two distinct creases with disjoint wing sets and the same target class, let their endpoint positions be $a_0 < a_1$ and $b_0 < b_1$. The pair is invalid exactly when the endpoints strictly alternate,

$$a_0 < b_0 < a_1 < b_1 \quad \text{or} \quad b_0 < a_0 < b_1 < a_1.$$

Equivalently, the two endpoint intervals must be nested or disjoint. Creases sharing a face are not part of this disjoint-wing test.

Let $F(P, k)$ be the exact set of physical globally valid labelled layer-order states for the period-expanded map, represented by their strict total orders. The local flat-foldability requirement is not a fourth order predicate, a state coordinate, a pivot parameter, or an auxiliary choice: it constrains

the prescribed assignment before any candidate order is tested. Consequently, an assignment that fails local flat-foldability has no physical state. For a locally admissible assignment, candidate-order membership is tested by the three global order-validity clauses above. Faces remain labelled and no symmetry quotient is taken. Fold sequences, auxiliary certificates, pivot parameters, and computational witnesses are not states.

Definition 3.2 (Period-2 invariants). Use the sign convention $M = +1$, $V = -1$ and multiplication of signs. Define

$$\rho = h_0 h_1, \quad \tau = u_0 u_1, \quad \delta_i = h_0 h_1 u_i l_i = \rho u_i l_i \quad (i = 0, 1),$$

and put

$$m = n - 1 = 2k - 1.$$

Further coordinates are introduced only when they become useful: row pivots in Section 6, the MVMMMM block coordinate B_R in Section 7.2, and the reflected coordinate B_L in Section 8 after the physical seam reflection has been established.

4 Local reduction and exact labelled symmetries

Definition 3.1 separates assignment-level physical admissibility from candidate-order validity. The three displayed order conditions test a proposed labelled total order; the local degree-4 rule is instead a necessary physical condition on the prescribed crease assignment itself. The next lemma specializes that prerequisite to the period-two words. Because the local rule depends only on P , not on a proposed order, failure at either physical parity makes $F(P, k)$ empty before any global pivot or merge analysis begins; no new state coordinate or order predicate is introduced.

Lemma 4.1 (Period-2 local obstruction). *For every $P \in \{M, V\}^6$ and every $k \geq 2$,*

$$F(P, k) \neq \emptyset \implies \delta_0 = \delta_1 = -1.$$

Equivalently, if $\delta_0 \neq -1$ or $\delta_1 \neq -1$, then $F(P, k) = \emptyset$.

Proof. At the separator between columns c and $c + 1$, the ordinary degree-4 interior vertex is incident to H_c, H_{c+1}, U_c, L_c . The standard local flat-foldability rule [1, 2] requires an odd number of mountain creases. Under the sign convention this is equivalent to

$$H_c H_{c+1} U_c L_c = -1.$$

With $i = (c - 1) \bmod 2$, period expansion gives

$$h_i h_{1-i} u_i l_i = h_0 h_1 u_i l_i = \delta_i.$$

For $k \geq 2$, both separator parities occur physically; at the minimum width $k = 2$, the separators 1, 2, 3 already have parities 0, 1, 0. Thus both equations are necessary. $\square$

Corollary 4.2 (The 48/16 local split). *Exactly 16 of the 64 formal period words satisfy $\delta_0 = \delta_1 = -1$. The other 48 have empty state set for every $k \geq 2$.*

Proof. Choose h_0, h_1, u_0, u_1 arbitrarily. The two local equations then determine

$$l_i = -\rho u_i \quad (i = 0, 1)$$

uniquely. Hence $2^4 = 16$ words survive the local test. The literal 64-word table is only a finite coverage check and is not a premise of this argument or of the all- k classification. $\square$

The remaining proofs repeatedly move between period words. Because states are labelled, the required symmetry statements must be exact state-set bijections rather than statements “up to symmetry.”

Proposition 4.3 (Exact labelled symmetries). *Let $n = 2k$ be even. Define*

$$\begin{aligned}\phi_R(A_c) &= B_c, & \phi_R(B_c) &= A_c, \\ \phi_H(A_c) &= A_{n+1-c}, & \phi_H(B_c) &= B_{n+1-c}.\end{aligned}$$

For a total order Π , let $\phi(\Pi)$ denote elementwise relabelling and $\text{rev}(\Pi)$ literal reversal. The following transformations are exact labelled state-set bijections:

g	word action gP	total-order action T_g
C	$(-h_0, -h_1, -u_0, -u_1, -l_0, -l_1)$	$\text{rev}(\Pi)$
R	$(h_0, h_1, l_0, l_1, u_0, u_1)$	$\text{rev}(\phi_R(\Pi))$
H	$(h_1, h_0, u_0, u_1, l_0, l_1)$	$\text{rev}(\phi_H(\Pi))$
CR	$(-h_0, -h_1, -l_0, -l_1, -u_0, -u_1)$	$\phi_R(\Pi)$
CH	$(-h_1, -h_0, -u_0, -u_1, -l_0, -l_1)$	$\phi_H(\Pi)$

For every listed g ,

$$\Pi \in F(P, k) \iff T_g(\Pi) \in F(gP, k),$$

and therefore

$$F(gP, k) = T_g(F(P, k))$$

as literal labelled sets.

Under horizontal reflection, the physical seam between columns p and $p+1$ is sent to the seam between columns $n-p$ and $n-p+1$. Thus the seam coordinate transforms as

$$p \longmapsto n - p.$$

Proof. Each face map preserves crease incidence, disjointness, and folded target class. Complement C reverses every directed M/V precedence by complementing the crease labels. Row swap R and horizontal reflection H reverse directed precedences by flipping checkerboard orientation. Hence C, R, H require literal order reversal, while the two reversal mechanisms cancel in CR and CH . Strict endpoint alternation is preserved by bijective relabelling and by order reversal. Each transformation is involutive, giving both directions of the state-set equivalence. Appendix B records the literal crease and parity maps. $\square$

5 Pleat singleton regime

When $\tau = -1$, the two row directions alternate in the way that turns primitive precedences into a full chain. This makes uniqueness transparent, but uniqueness is not yet existence: the forced chain still has to satisfy every long-range Butterfly condition.

Theorem 5.1 (Pleat singleton states). *Assume $k \geq 2$, $\delta_0 = \delta_1 = -1$, and $\tau = -1$. For*

$$P_+ = MMMVVM,$$

put

$$Q_q = (A_{2q-1}, B_{2q-1}, B_{2q}, A_{2q}), \quad 1 \leq q \leq k,$$

and

$$\Omega_+(k) = Q_1 Q_2 \cdots Q_k.$$

For

$$P_- = MVMVMV,$$

put

$$\Omega_-(k) = (A_1, A_2, \dots, A_n, B_n, B_{n-1}, \dots, B_1).$$

Then

$$F(MMMVVM, k) = \{\Omega_+(k)\}, \quad F(MVMVMV, k) = \{\Omega_-(k)\}.$$

The other six locally admissible pleat words have the exact singleton state sets

P	$F(P, k)$
$MMVMMV$	$\{T_R(\Omega_+(k))\}$
$VVMVVM$	$\{T_{CR}(\Omega_+(k))\}$
$VVVMMV$	$\{T_C(\Omega_+(k))\}$
$VMMVMV$	$\{T_H(\Omega_-(k))\}$
$VMVMVM$	$\{T_C(\Omega_-(k))\}$
$MVVMVM$	$\{T_{CH}(\Omega_-(k))\}$

Proof. For P_+ , the primitive precedences contain

$$A_{2q-1} \prec B_{2q-1} \prec B_{2q} \prec A_{2q}$$

inside each block and $A_{2q} \prec A_{2q+1}$ between consecutive blocks. These edges form a directed Hamiltonian path through all $4k$ faces, so every precedence-compatible total order equals $\Omega_+(k)$.

For P_- , the relations

$$A_c \prec A_{c+1} \ (1 \leq c < n), \quad A_n \prec B_n, \quad B_{c+1} \prec B_c \ (1 \leq c < n)$$

form the Hamiltonian path $A_1, \dots, A_n, B_n, \dots, B_1$, and hence force $\Omega_-(k)$.

These Hamiltonian chains establish uniqueness only. Appendix C verifies full validity for arbitrary separation in all same-target families H/H , U/U , L/L , and U/L . Thus both forced orders exist as globally valid states. The exact symmetry proposition then transports the two singleton states to the remaining six pleat words. $\square$

6 Row-local pivot normal forms

The case $\tau = +1$ does not collapse to a single chain. Instead, each physical row has a rigid one-parameter family of admissible orders. The parameter is a **pivot**: the vertical seam joining the first and last faces of the row order. Once the two row pivots are fixed, the global problem becomes a compatibility-and-merge problem between two prescribed row chains.

The underlying uniform-sign one-dimensional family is known in the stamp-folding literature: for a strip with q uniformly assigned creases, Hull et al. obtain exactly q labelled states and a two-spiral description indexed by the adjacent extreme faces [8, Theorem 2.1 and the proof of Theorem 3.1]. Proposition 6.1 records the same row family in the explicit seam coordinate and orientation conventions needed below. Its purpose here is to provide a self-contained row-normal-form

interface for the two-dimensional compatibility problem, not to assert priority for the one-dimensional classification.

For one row of width $2k$, write its faces as $F_1, \dots, F_{2k}$ and set

$$O_i = F_{2i-1}, \quad D_i = F_{2i} \quad (1 \leq i \leq k).$$

The odd seam $2i - 1$ joins (O_i, D_i) and has target class E ; the even seam $2i$ joins (D_i, O_{i+1}) and has target class W .

For effective sign $+$, impose the row-local directed constraints

$$O_i \prec D_i \quad (1 \leq i \leq k), \quad O_{i+1} \prec D_i \quad (1 \leq i < k),$$

together with noncrossing of all same-target seam spans within the row. For sign $-$, reverse the directed inequalities. Let $R_{\text{loc}}^\sigma(k)$ denote the resulting set of row-local orders.

For $1 \leq p \leq 2k - 1$, define $R^+(k, p)$ by

$$R^+(k, 2s - 1) = (O_s, O_{s-1}, \dots, O_1, D_1, \dots, D_{s-1}, O_{s+1}, \dots, O_k, D_k, \dots, D_{s+1}, D_s),$$

and

$$R^+(k, 2s) = (O_{s+1}, \dots, O_k, D_k, \dots, D_{s+1}, O_s, \dots, O_1, D_1, \dots, D_s),$$

with empty blocks omitted. Set

$$R^-(k, p) = \text{rev}(R^+(k, p)).$$

Proposition 6.1 (Row-local pivot normal forms). *For every $k \geq 2$ and $\sigma \in \{+, -\}$,*

$$R_{\text{loc}}^\sigma(k) = \{R^\sigma(k, p) : 1 \leq p \leq 2k - 1\},$$

and the pivot p is unique. It is the physical seam whose incident faces are the first and last faces of the row order.

If $R_{\text{glob}}^{P,r}(k)$ denotes the row restrictions induced by globally valid states of a fixed word P , then

$$R_{\text{glob}}^{P,r}(k) \subseteq R_{\text{loc}}^\sigma(k).$$

No converse is asserted. A global state must restrict to one of these row-local normal forms, but a row-local pivot need not extend to a global state.

Proof. Appendix A proves the adjacent-span trichotomy, the normal-form relation word, uniqueness of the extreme-face seam, and the local/global scope boundary. $\square$

For illustration only, take $k = 3$ and the odd pivot $p = 3$ (so $s = 2$). The formula gives

$$R^+(3, 3) = (O_2, O_1, D_1, O_3, D_3, D_2) = (F_3, F_1, F_2, F_5, F_6, F_4).$$

The first and last faces are therefore F_3 and F_4 , exactly the two faces incident to seam 3. This example illustrates the pivot coordinate and is not a global-realizability argument: Proposition 6.1 still asserts only the inclusion $R_{\text{glob}} \subseteq R_{\text{loc}}$.

For later use, let W_p denote the column sequence of $R^+(k, p)$. Thus W_p records the row order with the row label suppressed; $A(W_p)$ and $B(W_p)$ mean the corresponding upper and lower labelled row orders.

Pivot coordinates under the exact symmetries. Now that pivots have been defined, Proposition 4.3 gives their literal transport. Complement C leaves the numerical pivot unchanged. Row exchange R and CR swap the ordered upper/lower pivots,

$$(a, b) \mapsto (b, a).$$

Horizontal reflection H and CH apply the physical seam reflection to each row,

$$(a, b) \mapsto (n - a, n - b).$$

These are coordinate consequences of the exact labelled state bijections; they do not assert global realizability of arbitrary row-local pivots.

Table 1 collects the exact symmetry actions, including the pivot consequences now that pivots have been defined; it summarizes Proposition 4.3 and does not replace its bijection proof.

Table 1: Exact symmetry actions, including pivot-coordinate consequences. This table is placed only after pivots have been defined.

Action	Word action	Face/order action	Pivot action
C	global M/V complement	face labels fixed; reverse total order	$(a,b) \rightarrow (a,b)$
R	swap U/L period pairs	swap $A_c \leftrightarrow B_c$; reverse total order	$(a,b) \rightarrow (b,a)$
H	swap h0,h1	$c \rightarrow n+1-c$ in each row; reverse total order	$(a,b) \rightarrow (n-a, n-b)$
CR	complement after row swap	swap $A_c \leftrightarrow B_c$; no order reversal	$(a,b) \rightarrow (b,a)$
CH	complement after horizontal reflection	$c \rightarrow n+1-c$; no order reversal	$(a,b) \rightarrow (n-a, n-b)$

7 Canonical constant-row families

The four six-letter canonical words used repeatedly below are collected in Table 2; the table is a reading key for the pleat and constant-row mechanisms, not an additional classification claim.

Table 2: Canonical representatives and their structural mechanisms.

Word	ρ	τ	Mechanism	Compatibility	Multiplicity
MMVMV	+1	-1	pleat Hamiltonian precedence + full Butterfly audit	N/A	one exact labelled state
MVMVM	-1	-1	pleat row-monotone/reverse-row chain + full Butterfly audit	N/A	one exact labelled state
MMVMV	+1	+1	row-local pivots + global merge	$a=1$ OR $a=m$ OR $a=b$	one labelled state per compatible ordered pair
MVMMM	-1	+1	LIFO horizontal reconstruction + pivot blocks	$B_R(a)=B_R(b)$	one labelled state per compatible ordered pair

The local reduction and symmetries leave two essentially different constant-row mechanisms. In both, the pivot records the only possible row-local order. What remains is to decide which ordered pivot pairs can be merged into a global state and whether such a merge is unique.

7.1 The canonical word $MMVMV$

For $P = MMVMV$, both physical rows use the + normal form. A pivot pair (a, b) therefore prescribes the two row chains $A(W_a)$ and $B(W_b)$. Let $\text{FullCompat}(k; a, b)$ mean that at least one globally valid labelled state has exactly these row restrictions.

The criterion has a simple shape: every diagonal pair works, and every pair with the upper pivot at either boundary works. The proof separates obstruction, construction, full validity, and uniqueness; in particular, local legality of the next horizontal event is not the same as existence of a completable merge.

Theorem 7.1 (Canonical MMMMVV compatibility and multiplicity one). *Let $k \geq 2$, $m = 2k - 1$, and $1 \leq a, b \leq m$. Then*

$$\text{FullCompat}(k; a, b) \iff (a = 1) \text{ or } (a = m) \text{ or } (a = b).$$

Every compatible ordered pivot pair determines exactly one globally valid labelled state. Conversely, every globally valid labelled MMMMVV state determines a unique ordered pivot pair.

Proof. 1. Interior off-diagonal obstruction. Write $o_r = 2r - 1$ and $e_r = 2r$. For a row word W_p , define

$$X_p(i, j) \iff e_j \text{ precedes } o_i \text{ in } W_p.$$

The normal-form formulas give

$$X_{2s-1}(i, j) \iff i > s \text{ and } j < s,$$

$$X_{2s}(i, j) \iff i \leq s \text{ and } j > s.$$

If $1 < a < m$ and $a \neq b$, Appendix D.2 supplies a parity-complete witness with $X_a(i, j) = 1$ and $X_b(i, j) = 0$. With $o = 2i - 1$ and $e = 2j$, the two row orders and the horizontal MMMMVV precedences force the four-cycle

$$A_e \prec A_o \prec B_o \prec B_e \prec A_e,$$

so no global state exists.

2. Allowed constructions. Appendix D.3 gives five explicit families: the diagonal construction and four boundary constructions. Their pivot pairs cover exactly $a = b$, $a = 1$, and $a = m$, including all endpoint cases at $k = 2$.

3. Full validity. Each construction has the required row restrictions and a laminar horizontal stack; Appendix D.4 records the horizontal-stack check. Row-local validity supplies all within-row vertical conditions. Appendix D.5 discharges the remaining equal-parity U_i/L_j Butterfly family at arbitrary separation. It first relates lower-face insertion slots to actual endpoint alternation, then uses the constant endpoint-sum structure of the four boundary shell families; the diagonal construction is handled separately through its contiguous column blocks and row-local noncrossing.

4. Fixed-pivot uniqueness. A partial horizontal merge can admit more than one H-locally legal next event even when only one choice has a complete H-valid continuation, so the proof never identifies local legality with completability. For diagonal pairs $a = b$, Lemma D.2 shows directly that the matching closer is forced after each opener and hence that $D(k, a)$ is the unique complete H-valid merge. For boundary pairs $a = 1$ or $a = m$, Appendix D.7 compares any hypothetical second complete merge with the explicit construction from D.3–D.4 and takes their first divergence. Projected determinism leaves only an upper-odd versus lower-even opener choice. Appendix D.8 lists exactly such windows along the known complete constructions, and Appendix D.9 excludes the alternative branch by directed requirement cycles. The H-local/H-completability distinction is isolated in Appendix D.6. Boundary and diagonal pairs exhaust the allowed set, so every allowed pivot pair has at most one full state, while the explicit constructions supply one.

5. State–pivot bijection. Every global state restricts to two row-local normal forms, so Proposition 6.1 assigns it a unique ordered pivot pair. The obstruction restricts the image to the compatible pairs; existence and fixed-pair uniqueness give the inverse. Hence globally valid states and compatible ordered pivot pairs are in bijection; Appendix D.10 records this bridge in the technical closure. $\square$

ordered pivot pairs (a, b) , schematic $m = 7$

	1	2	3	4	5	6	7
1	●	●	●	●	●	●	●
2		●					
3			●				
4				●			
5					●		
6						●	
7	●	●	●	●	●	●	●

b

filled cells: $a = 1$ or $a = m$ or $a = b$

Figure 2: Ordered-pair compatibility matrix for canonical $MMMMVV$, shown schematically at $m = 7$. The horizontal coordinate is the lower-row pivot b and the vertical coordinate is the upper-row pivot a . A filled cell is compatible exactly when $a = 1$, $a = m$, or $a = b$; the two boundary conditions are therefore explicitly asymmetric in a .

7.2 The canonical word $MVMMMM$

For $P = MVMMMM$, the upper row uses $R^+(k, a)$ and the lower row uses $R^-(k, b)$. Here the opposite row orientations make the horizontal system especially rigid: upper faces open horizontal intervals, lower faces close them, and H/H noncrossing becomes a LIFO reconstruction problem.

Call a merge **H-compatible** if it satisfies all horizontal precedences and all H/H Butterfly constraints. Let $\text{HCompat}(k; a, b)$ and $\text{FullCompat}(k; a, b)$ denote existence of an H-compatible and a globally valid merge, respectively.

Define the right block coordinate

$$B_R(1) = 0, \quad B_R(p) = \left\lfloor \frac{p}{2} \right\rfloor \quad (p \geq 2).$$

Thus

$$Q_0 = \{1\}, \quad Q_r = \{2r, 2r + 1\} \quad (1 \leq r \leq k - 1).$$

The coordinate B_R simply records which of these pivot blocks contains p .

Theorem 7.2 (Canonical $MVMMMM$ block compatibility and multiplicity one). *For $k \geq 2$ and $1 \leq a, b \leq m = 2k - 1$,*

$$\text{FullCompat}(k; a, b) \iff B_R(a) = B_R(b).$$

For every compatible ordered pivot pair there is exactly one globally valid labelled state, and every globally valid state determines one such pair through its unique row-normal-form pivots.

Proof. The proof has two independent steps.

Step A: horizontal reconstruction is enough. In $MVMMMM$, every horizontal precedence is upper-before-lower. Therefore upper faces push their horizontal labels and lower faces pop them. Pairwise H/H noncrossing is exactly the LIFO stack condition. With the two row chains fixed, the next requested closer is fixed: if its opener has not yet appeared, the upper chain is forced until it appears; if the label is on top of the stack, its closer is forced; if it is buried, completion is impossible. Hence an H-compatible merge, when it exists, is unique.

Successful H reconstruction is already globally valid. Proposition 6.1 supplies all within-row vertical precedences and same-row vertical noncrossing. The only remaining same-target class is U_i/L_j with $i \equiv j \pmod{2}$. Appendix E proves the side-crossing transfer: a crossing of such a cross-row pair would force a crossing in the corresponding upper or lower same-row pair. Those are forbidden by the row-normal forms. Thus

$$\text{FullCompat}(k; a, b) \iff \text{HCompat}(k; a, b).$$

Step B: H compatibility is a block condition. For $1 \leq r \leq k - 1$, write the two pivots in block r as

$$p_r^{(0)} = 2r, \quad p_r^{(1)} = 2r + 1.$$

Let $c_r = 2r + 1$, $d_r = 2r + 2$, and define

$$L_r = (2r - 1, 2r - 3, \dots, 1, 2, 4, \dots, 2r),$$

$$R_r = (2r + 3, 2r + 5, \dots, 2k - 1, 2k, 2k - 2, \dots, 2r + 4).$$

The row words have the endpoint normal forms

$$W_{p_r^{(0)}} = c_r R_r d_r L_r, \quad W_{p_r^{(1)}} = c_r L_r R_r d_r.$$

Appendix E proves that unequal blocks always contain a three-label LIFO obstruction, whereas each of the four ordered pivot pairs inside one nonzero block has an explicit legal push/pop schedule; the singleton block Q_0 uses push-all/pop-all. Consequently

$$\text{HCompat}(k; a, b) \iff B_R(a) = B_R(b).$$

Combining Steps A and B gives the criterion. Step A gives fixed-pair uniqueness, and Proposition 6.1 gives unique pivots for every global state, so states and compatible ordered pivot pairs are in bijection. $\square$

ordered pivot pairs (a, b) , schematic $m = 7$

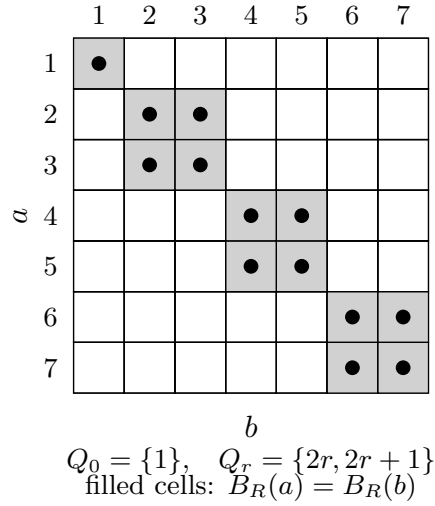


Figure 3: Ordered-pair block compatibility matrix for canonical $MVMMM$, shown schematically at $m = 7$. The axes are the ordered pivots (a, b) . The singleton block is $Q_0 = \{1\}$ and the nonzero blocks are $Q_r = \{2r, 2r+1\}$. A filled cell satisfies $B_R(a) = B_R(b)$; each nonzero 2×2 block visibly contains four distinct ordered compatible pairs.

8 Exact transports and the all-64 classification

The two canonical constant-row families are now completely understood. Exact labelled symmetries transfer them to the remaining constant-row words, after which the all-64 theorem is only a synthesis of the local split and the three nonzero structural mechanisms.

Horizontal reflection has already been shown to send a physical seam by $p \mapsto n - p$. Only now define the reflected block coordinate

$$B_L(p) := B_R(n - p), \quad 1 \leq p \leq m.$$

Proposition 8.1 (Exact transports of the constant-row fibres). *Assume $k \geq 2$, $\delta_0 = \delta_1 = -1$, and $\tau = +1$.*

If $\rho = +1$, the four words are

$$MMMMVV, \quad MMVVMM, \quad VVMMVV, \quad VVVVMM,$$

and their exact state sets are respectively

$$F(MMMM VV, k), \quad T_R(F(MMMM VV, k)), \\ T_{CR}(F(MMMM VV, k)), \quad T_C(F(MMMM VV, k)).$$

Writing (α, β) for the target upper/lower pivots, the target compatibility criterion is

$$\alpha = 1 \text{ or } \alpha = m \text{ or } \alpha = \beta$$

for the canonical and complement targets, and

$$\beta = 1 \text{ or } \beta = m \text{ or } \alpha = \beta$$

for the row-swapped targets, because R and CR exchange the ordered pivots.

If $\rho = -1$, the four words are

$$MVMMMM, \quad VM MMMM, \quad VMVVVV, \quad MVVVVV,$$

and their exact state sets are respectively

$$F(MVMMMM, k), \quad T_H(F(MVMMMM, k)), \\ T_C(F(MVMMMM, k)), \quad T_{CH}(F(MVMMMM, k)).$$

For the canonical and complement targets, compatibility is $B_R(\alpha) = B_R(\beta)$. For the horizontally reflected targets, the source pivots first transform geometrically as

$$(\alpha, \beta) = (n - a, n - b),$$

and only then the target criterion is expressed as

$$B_L(\alpha) = B_L(\beta).$$

Every compatible target pivot fibre is a singleton.

Proof. Proposition 4.3 gives literal state-set bijections, and the pivot consequences recorded after Proposition 6.1 give the exact coordinate maps. Apply these bijections to the two canonical theorems. Since both the state map and pivot map are bijections, each target pivot fibre is the image of exactly one canonical fibre, so multiplicity one transports fibrewise. No count equality is used. $\square$

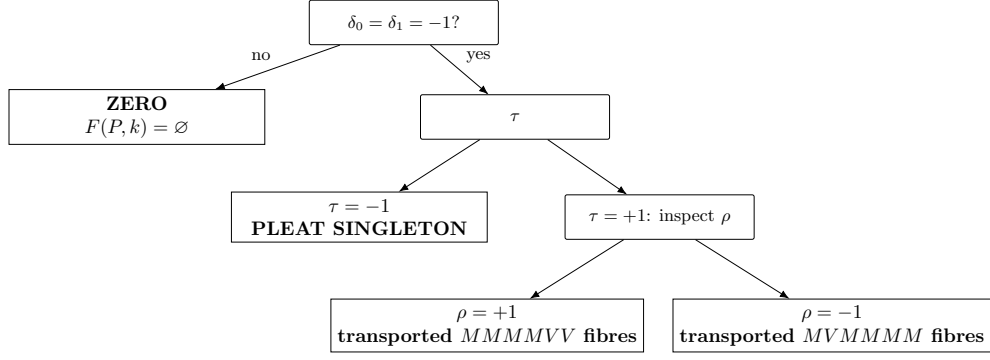


Figure 4: Decision diagram for the exact state-set classification. Local admissibility is tested first by δ_0 and δ_1 ; τ separates the pleat and constant-row regimes, and ρ separates the two constant-row canonical families. Counts are deliberately omitted because exact labelled state sets are primary.

Theorem 8.2 (Exact labelled classification of all period-2 words). *Let $P \in \{M, V\}^6$ and $k \geq 2$. The exact labelled state set $F(P, k)$ is determined by exactly one of the following four regimes.*

regime	condition	exact state set
Zero	$\delta_0 \neq -1$ or $\delta_1 \neq -1$	$F(P, k) = \emptyset$
Pleat singleton	$\delta_0 = \delta_1 = -1, \tau = -1$	the exact singleton from Theorem 5.1, transported by the labelled symmetries
Constant row, $\rho = +1$	$\delta_0 = \delta_1 = -1, \tau = +1, \rho = +1$	the exact transported MMMMVV fibres of Proposition 8.1, using the canonical criterion of Theorem 7.1; each compatible fibre is a singleton
Constant row, $\rho = -1$	$\delta_0 = \delta_1 = -1, \tau = +1, \rho = -1$	the exact transported MVMMMM fibres of Proposition 8.1, using the canonical criterion of Theorem 7.2; each compatible fibre is a singleton

In particular, the theorem classifies the literal labelled state sets, not only their cardinalities.

Proof. Lemma 4.1 excludes the 48 locally obstructed words, and Corollary 4.2 leaves exactly 16 locally admissible words. Among those, $\tau = -1$ gives exactly the eight pleat words handled in Section 5. If $\tau = +1$, the sign ρ separates the remaining eight words into the two groups of four transported from MMMMVV and MVMMMM. Proposition 8.1 gives the exact state-set transport in each group. The alternatives are mutually exclusive and exhaustive, so all 64 formal words are covered. The literal 64-word table is a finite cross-check only. $\square$

Table 3: Invariant/regime summary. The exact state-set description precedes the downstream count.

Local condition	τ	ρ	Type	Exact state set	Count
either δ_0 or $\delta_1 \neq -1$	-	-	Zero	empty	0
$\delta_0 = \delta_1 = -1$	-1	either	Pleat singleton	exact singleton transported from one of two canonical pleat orders	1
$\delta_0 = \delta_1 = -1$	+1	+1	Constant row: MMMMVV	exact transported singleton fibres indexed by compatible ordered pivots	6k-5
$\delta_0 = \delta_1 = -1$	+1	-1	Constant row: MVMMMM	exact transported singleton fibres indexed by equal block coordinate	4k-3

Corollary 8.3 (Closed counts). *Define, downstream of the exact state-set theorem,*

$$C_P(k) := |F(P, k)|.$$

For every $k \geq 2$,

$$C_P(k) = \begin{cases} 0, & \delta_0 \neq -1 \text{ or } \delta_1 \neq -1, \\ 1, & \delta_0 = \delta_1 = -1, \tau = -1, \\ 6k - 5, & \delta_0 = \delta_1 = -1, \tau = +1, \rho = +1, \\ 4k - 3, & \delta_0 = \delta_1 = -1, \tau = +1, \rho = -1. \end{cases}$$

Proof. The zero and pleat values are immediate from the exact state sets. For $MMMMVV$, compatible pivot pairs are

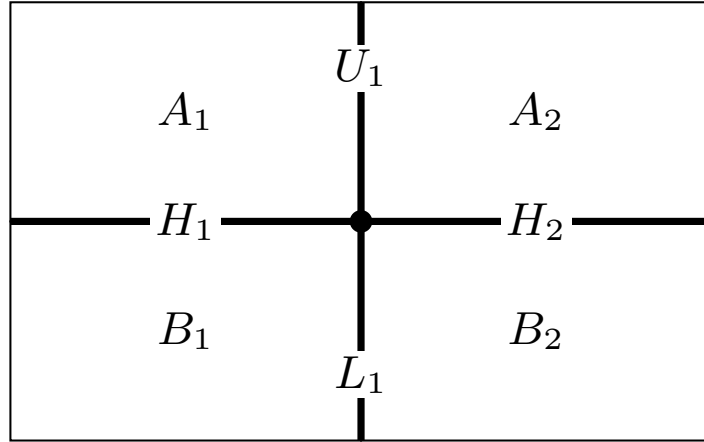
$$\{(1, b) : 1 \leq b \leq m\} \cup \{(m, b) : 1 \leq b \leq m\} \cup \{(a, a) : 1 \leq a \leq m\},$$

with only $(1, 1)$ and (m, m) counted twice, so there are $3m - 2 = 6k - 5$ singleton fibres. For $MVMMMM$, $Q_0 = \{1\}$ contributes one ordered pair and each of the $k - 1$ two-element blocks contributes four, giving $1 + 4(k - 1) = 4k - 3$. Exact transports preserve these cardinalities. No count enters the proof of Theorem 8.2. $\square$

9 Boundary case $k = 1$

The width-2 map is a separate physical branch. Only the creases H_1, H_2, U_1, L_1 exist; the formal bits u_1, l_1 have no physical carriers. Accordingly the generic two-parity argument, the row-pivot machinery, and the $k \geq 2$ classification are not used here.

$k = 1, n = 2$ boundary geometry



physical creases: H_1, H_2, U_1, L_1 only

formal bits u_1, l_1 have no physical carriers

Figure 5: Width-two boundary geometry for $k = 1$. The physical map has four faces and exactly four creases H_1, H_2, U_1, L_1 meeting at the central degree-4 vertex. The formal period bits u_1 and l_1 have no physical carriers and therefore do not enter the $k = 1$ criterion.

Proposition 9.1 (Exact boundary state set at $k = 1$). *Let*

$$P = (h_0, h_1, u_0, u_1, l_0, l_1) \in \{M, V\}^6$$

and define the physical projection

$$\pi_1(P) = (h_0, h_1, u_0, l_0).$$

Formal words with the same $\pi_1(P)$ define the same physical map and the same labelled state set. Put

$$\delta_0(P) = h_0 h_1 u_0 l_0.$$

Then

$$\delta_0 = +1 \implies F(P, 1) = \emptyset,$$

whereas

$$\delta_0 = -1 \implies F(P, 1) = \{\Omega(\pi_1(P))\}$$

for one exact labelled total order $\Omega(\pi_1(P))$.

Proof. The four physical creases form the incidence cycle

$$A_1 - H_1 - B_1 - L_1 - B_2 - H_2 - A_2 - U_1 - A_1.$$

Orient its edges by primitive M/V precedence. Relative to this cyclic traversal, the product of the four orientation signs is δ_0 . When $\delta_0 = -1$, exactly one or three edges oppose the traversal; the three majority-oriented consecutive edges form a directed Hamiltonian path through all four faces, and the remaining edge is a source-to-sink shortcut. Hence there is one linear extension. Its two opposite-edge Butterfly pairs are respectively disjoint and nested, so that order is globally valid.

When $\delta_0 = +1$, local flat-foldability fails. Appendix F also gives the direct global-semantic obstruction: 0 or 4 traversal reversals produce a directed cycle, while every 2-reversal orientation forces strict alternation in one opposite-edge pair. $\square$

Consequently

$$|F(P, 1)| = 1 \iff \delta_0 = -1, \quad |F(P, 1)| = 0 \iff \delta_0 = +1.$$

Appendix F lists the eight admissible physical projections and their exact singleton orders.

10 Finite validation and reproducibility

The finite computations provide independent validation checks; the all- k classification is established by the symbolic arguments in the preceding sections. The independent finite validation compares exact labelled state sets, rather than counts alone, so that it can detect errors in labels, transports, pivot recovery, or noncrossing semantics that happen to preserve cardinality.

At $k = 2$ the finite verification covers all 64 formal period words, the exact pivot fibres of the canonical $MMMMVV$ and $MVMMMM$ families, and all five exact symmetry actions. A separate exhaustive check covers all 64 formal words at $k = 1$ and verifies invariance under the physically absent u_1, l_1 bits. The literal all-word tables are supplied in Supplement S1, bounded exact validation records in Supplement S2, and the clean generation scripts, environment record, and checksums in Supplement S3.

The publication reproducibility package regenerates the support tables and figures and reruns the bounded semantic checks without requiring the historical development archive. These computational products therefore provide independent bounded verification of the presentation and implementation, but no finite dataset is used to infer an infinite-family statement.

11 Discussion and scope

The classification concerns ordinary orthogonal $2 \times 2k$ maps with a six-bit period-two formal assignment and strict total orders of labelled faces. No symmetry quotient is taken, and distinct fold sequences that end in the same labelled total order are not distinguished. The theorem therefore does not classify arbitrary-period assignments, other map geometries, fold-sequence multiplicity, or unlabelled symmetry orbits.

At the state-set level, the two constant-row regimes have genuinely different structural descriptions: $MMMMVV$ is controlled by a boundary-or-diagonal compatibility condition, while $MVMMMM$ is controlled by equality of the block coordinate B_R . Multiplicity one converts those compatibility sets directly into exact labelled fibres, which is why the count formulas can remain secondary to the structural theorem. The separate $k = 1$ proposition reflects a physical change in the crease set rather than an exception obtained by substituting $k = 1$ into the generic two-parity argument.

The supplementary computation provides exact bounded validation checks, literal classification tables, and a clean reproduction path for the supplementary figures and tables. These finite products are useful for detecting transcription or implementation errors, but they do not enlarge the scope of the symbolic theorems.

12 Conclusion

The period-two $2 \times 2k$ family admits an exact labelled state-set description built from a local obstruction, exact transports, a pleat singleton mechanism, and two canonical constant-row compatibility mechanisms. The contrast between the constant-row branches is structural: $MMMMVV$ allows boundary-or-diagonal pivot pairs, whereas $MVMMMM$ allows exactly equal block-coordinate pairs. Compatible ordered pivot pairs parameterize labelled states with multiplicity one, and the formulas $6k - 5$ and $4k - 3$ follow by counting those fibres. The width-two boundary has its own empty-or-singleton criterion because its physical crease set omits two formal period bits.

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A Row-Local Normal Forms and the Local/Global Scope Boundary

This appendix proves Proposition 6.1 using only row-local directed precedences and same-target vertical noncrossing.

A.1 Adjacent-span trichotomy

For effective sign $+$, put $J_i = (O_i, D_i)$. For adjacent J_i, J_{i+1} , the directed inequalities are $O_i \prec D_i$, $O_{i+1} \prec D_{i+1}$, and $O_{i+1} \prec D_i$. Of the six total orders compatible with the first two inequalities, one disjoint order contradicts $O_{i+1} \prec D_i$ and the two alternating orders are forbidden by Butterfly noncrossing. The only possibilities are

$$L_i : O_{i+1} \prec O_i \prec D_i \prec D_{i+1},$$

$$R_i : O_i \prec O_{i+1} \prec D_{i+1} \prec D_i,$$

$$C_i : O_{i+1} \prec D_{i+1} \prec O_i \prec D_i.$$

A.2 Relation-word normal form

Let $K_i = (O_{i+1}, D_i)$ be the even-seam span. For consecutive relation symbols, each of $(R, L), (R, C), (C, L), (C, C)$ forces strict alternation of K_i and K_{i+1} and is therefore impossible. Hence every admissible relation word is exactly

$$L^*R^* \quad \text{or} \quad L^*CR^*.$$

A relation word $L_1 \cdots L_{s-1} R_s \cdots R_{k-1}$ forces two nested chains around O_s, D_s . At an interior junction, putting O_{s+1} before D_{s-1} would make K_{s-1} and K_s alternate, so $D_{s-1} \prec O_{s+1}$. This fixes the entire order as $R^+(k, 2s-1)$.

A relation word $L_1 \cdots L_{s-1} C_s R_{s+1} \cdots R_{k-1}$ has the right block wholly before the left block because C_s gives $D_{s+1} \prec O_s$. This fixes the order as $R^+(k, 2s)$.

Thus every row-local $+$ order is one of the explicit normal forms.

A.3 Sufficiency of the normal forms

In each odd-pivot normal form, all $O_i \prec D_i$ and $O_{i+1} \prec D_i$ inequalities are immediate from the two monotone blocks around the central pair. The J spans form two nested chains, mutually disjoint except for the central containing span; the K spans similarly form two nested chains. Hence every required same-target pair is noncrossing.

For an even pivot, the right block precedes the left block. The directed inequalities again hold, including the central even-seam inequality. The J spans split into two nested chains separated by the central C relation. The span K_s contains the whole order, while the remaining K spans form two nested chains inside it. Thus all row-local constraints hold.

Therefore

$$R_{\text{loc}}^+(k) = \{R^+(k, p) : 1 \leq p \leq 2k-1\}.$$

Literal reversal reverses every directed inequality and preserves endpoint alternation, giving

$$R_{\text{loc}}^-(k) = \{R^-(k, p) : 1 \leq p \leq 2k-1\}.$$

A.4 Pivot uniqueness

For $p = 2s-1$, the first and last faces are O_s and D_s , the wings of seam $2s-1$. For $p = 2s$, they are O_{s+1} and D_s , the wings of seam $2s$. Hence the extreme pair identifies one physical seam and therefore one pivot. Reversal exchanges first and last but preserves the same seam.

A.5 Local/global firewall

Every global state restricts to a row order satisfying the row-local directed and Butterfly constraints. Therefore

$$R_{\text{glob}}^{P,r}(k) \subseteq R_{\text{loc}}^\sigma(k).$$

No reverse inclusion follows from the argument above. In particular, the row-local count $2k-1$ does not imply that all $2k-1$ pivots occur globally for an arbitrary period word. Global realization is supplied only by the downstream canonical compatibility theorem in a family where it has actually been proved.

B Exact labelled symmetry maps

This appendix gives the literal semantic verification behind Proposition 4.3. Pivot coordinates are treated only in B.6, after the row-normal-form pivot has been defined in Section 6.

B.1 Complement C

Complement fixes every face and crease while replacing every M/V label by its opposite. Checkerboard orientation and target classes are unchanged, so every primitive directed precedence reverses. Literal reversal of the total order reverses every strict inequality and preserves strict endpoint alternation. Hence

$$T_C(\Pi) = \text{rev}(\Pi)$$

is an exact labelled state bijection.

B.2 Row swap R

The face map ϕ_R exchanges A_c and B_c , maps U_i to L_i and conversely, and fixes H_c . It flips checkerboard light/dark orientation while preserving the M/V labels. Seam index, and therefore the E/W target class, is unchanged. Thus primitive precedences reverse and

$$T_R(\Pi) = \text{rev}(\phi_R(\Pi)).$$

Incidence, disjointness, target class, and strict-alternation status are preserved.

B.3 Horizontal reflection H

The map ϕ_H sends columns by $c \mapsto n + 1 - c$, so

$$H_c \mapsto H_{n+1-c}, \quad U_i \mapsto U_{n-i}, \quad L_i \mapsto L_{n-i}.$$

Because n is even, the checkerboard orientation of every face flips. Horizontal-crease period parity therefore swaps h_0 and h_1 . By contrast, $n - i$ has the same parity as i , so the u and l period slots and the E/W target classes are preserved. The precedence reversal caused by the checkerboard flip gives

$$T_H(\Pi) = \text{rev}(\phi_H(\Pi)).$$

A physical seam p between columns p and $p + 1$ maps to the seam between columns $n - p$ and $n - p + 1$; therefore

$$p \mapsto n - p.$$

B.4 The compositions CR and CH

For CR , row swap reverses the directed precedences and complement reverses them again. The two effects cancel, hence

$$T_{CR}(\Pi) = \phi_R(\Pi).$$

For CH , horizontal reflection and complement likewise contribute two reversals, so

$$T_{CH}(\Pi) = \phi_H(\Pi).$$

In both cases the same incidence, target-class, and Butterfly-preservation arguments apply.

B.5 Exact two-direction state-set equivalence

Each face transformation is bijective and preserves crease incidence and disjointness. Corresponding creases have the same folded target class. Bijective relabelling preserves endpoint positions up to renaming, and literal order reversal is an order anti-isomorphism; both preserve whether two disjoint same-target crease intervals strictly alternate. Thus every semantic clause holds for Π exactly when it holds for $T_g(\Pi)$. Since every listed transformation is involutive,

$$\Pi \in F(P, k) \iff T_g(\Pi) \in F(gP, k),$$

and therefore

$$F(gP, k) = T_g(F(P, k)).$$

This is equality of literal labelled state sets, not a symmetry quotient.

B.6 Pivot coordinates after the row-normal-form definition

Let $R^\sigma(k, p)$ be the row-local normal form of Proposition 6.1, where p is the unique physical seam joining the extreme faces of the row order. Then

$$T_C(R^\sigma(k, p)) = R^{-\sigma}(k, p),$$

so complement leaves the numerical pivot unchanged.

Row swap exchanges physical rows without reflecting seam indices. Thus on an ordered upper/lower pair,

$$(a, b) \mapsto (b, a)$$

under both R and CR .

Horizontal reflection sends the physical seam p to $n - p$. The row normal forms satisfy

$$T_H(R^\sigma(k, p)) = R^\sigma(k, n - p),$$

$$T_{CH}(R^\sigma(k, p)) = R^{-\sigma}(k, n - p),$$

and therefore

$$(a, b) \mapsto (n - a, n - b)$$

under H and CH . These identities transport row-local pivots; they do not assert that an arbitrary local pivot is globally realizable.

C Full Validity of the Canonical Pleat Orders

The Hamiltonian arguments in Theorem 5.1 prove uniqueness among precedence-compatible orders. This appendix proves existence by checking every Butterfly family at arbitrary separation.

C.1 The order $\Omega_+(k)$

For $1 \leq q \leq k$ the positions are

$$\text{pos}(A_{2q-1}) = 4q - 3, \quad \text{pos}(B_{2q-1}) = 4q - 2,$$

$$\text{pos}(B_{2q}) = 4q - 1, \quad \text{pos}(A_{2q}) = 4q.$$

All primitive M/V precedences point forward in this order. The sorted endpoint intervals are

$$I(H_{2q-1}) = [4q - 3, 4q - 2], \quad I(H_{2q}) = [4q - 1, 4q],$$

$$I(U_{2q-1}) = [4q - 3, 4q], \quad I(U_{2q}) = [4q, 4q + 1],$$

$$I(L_{2q-1}) = [4q - 2, 4q - 1], \quad I(L_{2q}) = [4q - 1, 4q + 2],$$

where the even-seam formulas run only to $q = k - 1$.

For H/H , distinct column intervals are strictly disjoint. For same-parity U/U and L/L , if $q < r$ the right endpoint of the q interval is strictly before the left endpoint of the r interval; hence these pairs are disjoint.

For the cross-row U/L class there is no distance cutoff. For odd seams $i = 2q - 1$, $j = 2r - 1$,

$$I(U_i) = [4q - 3, 4q], \quad I(L_j) = [4r - 2, 4r - 1].$$

If $r = q$, the lower interval is strictly nested in the upper one; if $r < q$ it is strictly before it; if $r > q$ it is strictly after it. For even seams $i = 2q$, $j = 2r$,

$$I(U_i) = [4q, 4q + 1], \quad I(L_j) = [4r - 1, 4r + 2],$$

and the same trichotomy holds, with the upper interval nested in the lower one when $r = q$. Thus every same-target pair is nested or disjoint for arbitrary q, r .

C.2 The order $\Omega_-(k)$

Here

$$\text{pos}(A_c) = c, \quad \text{pos}(B_c) = 2n - c + 1.$$

The primitive directed relations are $A_i \prec A_{i+1}$, $B_{i+1} \prec B_i$, and $A_c \prec B_c$, all of which point forward.

For horizontal creases,

$$I(H_c) = [c, 2n - c + 1].$$

If $c < d$, then

$$c < d < 2n - d + 1 < 2n - c + 1,$$

so $I(H_d)$ is strictly nested in $I(H_c)$.

Upper vertical intervals are $I(U_i) = [i, i + 1]$. Distinct same-target seams have equal parity and hence differ by at least two, so these intervals are disjoint. Lower vertical intervals are

$$I(L_i) = [2n - i, 2n - i + 1]$$

and are likewise disjoint within a target parity.

Finally,

$$\max I(U_i) \leq n < n + 1 \leq \min I(L_j)$$

for all i, j , so every cross-row U/L pair is separated, even before target parity is imposed.

C.3 Boundary and strictness check

The Butterfly predicate uses strict alternation. Genuine tested pairs have disjoint wing sets, so their four endpoint positions are distinct. At the left and right boundaries all displayed interval formulas stay within $1, \dots, 4k$; even seam formulas stop at $n - 2$. Therefore no endpoint degeneration or omitted long-range case occurs. The families H/H , U/U , L/L , and U/L exhaust all same-target disjoint crease pairs because horizontal target S is distinct from vertical targets E, W .

Hence both canonical pleat orders satisfy every global semantic condition for every $k \geq 2$.

D Technical Closure for $MMMMVV$

This appendix supplies the detailed necessity, existence, full-validity, and uniqueness bridges used in Theorem 7.1.

D.1 Row-normal-form column words and horizontal actions

Put $o_r = 2r - 1$, $e_r = 2r$ for $1 \leq r \leq k$. The plus-normal-form column word is

$$W_{2s-1} = (o_s, o_{s-1}, \dots, o_1, e_1, \dots, e_{s-1}, o_{s+1}, \dots, o_k, e_k, \dots, e_{s+1}, e_s),$$

$$W_{2s} = (o_{s+1}, \dots, o_k, e_k, \dots, e_{s+1}, o_s, \dots, o_1, e_1, \dots, e_s).$$

For $MMMMVV$, horizontal crease H_{o_r} has opener A_{o_r} and closer B_{o_r} , while H_{e_r} has opener B_{e_r} and closer A_{e_r} .

D.2 Necessity: the four-edge cycle

Define $X_p(i, j)$ by e_j preceding o_i in W_p . Direct inspection gives

$$X_{2s-1}(i, j) \iff (i > s \text{ and } j < s),$$

$$X_{2s}(i, j) \iff (i \leq s \text{ and } j > s).$$

If $1 < a < m$ and $a \neq b$, choose

$$(i, j) = (s + 1, s - 1) \quad \text{when } a = 2s - 1,$$

$$(i, j) = (s, s + 1) \quad \text{when } a = 2s.$$

The legal ranges are $2 \leq s \leq k - 1$ in the odd case and $1 \leq s \leq k - 1$ in the even case. The following complete case table verifies $X_a = 1$ and $X_b = 0$:

parity of (a, b)	condition	witness	reason $X_b = 0$
odd/odd	$a < b$	$(s + 1, s - 1)$	X_b would require $i > t$, but $t \geq s + 1 = i$
odd/odd	$a > b$	$(s + 1, s - 1)$	X_b would require $j < t$, but $t \leq s - 1 = j$

parity of (a, b)	condition	witness	reason $X_b = 0$
odd/even	any $b \neq a$	$(s+1, s-1)$	if $b = 2t$, then $X_b = 1$ would require $s+1 \leq t$ and $s-1 > t$, simultaneously
even/odd	any $b \neq a$	$(s, s+1)$	if $b = 2t-1$, then $X_b = 1$ would require $s > t$ and $s+1 < t$, simultaneously
even/even	$a < b$	$(s, s+1)$	X_b would require $j > t$, but $t \geq s+1 = j$
even/even	$a > b$	$(s, s+1)$	X_b would require $i \leq t$, but $t \leq s-1 < i = s$

Set $o = 2i - 1$ and $e = 2j$. In any merge with pivots (a, b) ,

$$A_e \prec A_o$$

comes from $X_a = 1$ in the upper row. Since o is odd and e even, horizontal orientation gives

$$A_o \prec B_o, \quad B_e \prec A_e.$$

Since $X_b = 0$, the strict lower row gives $B_o \prec B_e$. Thus

$$A_e \prec A_o \prec B_o \prec B_e \prec A_e,$$

impossible in a total order. At $k = 2$, the only interior upper pivot is $a = 2$, the witness is $(i, j) = (1, 2)$, and the same cycle is $A_4 \prec A_1 \prec B_1 \prec B_4 \prec A_4$.

D.3 Five explicit existence families

For a column c , define

$$Q_c = (A_c, B_c) \quad (c \text{ odd}), \quad Q_c = (B_c, A_c) \quad (c \text{ even}).$$

Empty ranges are omitted.

If $W_p = (c_1, \dots, c_{2k})$, define the diagonal construction

$$D(k, p) = Q_{c_1} Q_{c_2} \cdots Q_{c_{2k}},$$

which has pivots (p, p) .

For $1 \leq s \leq k$ define

$$L_{\text{odd}}(k, s) = A_{o_1} \cdots A_{o_s} \mid B_{o_s} \cdots B_{o_1} \mid B_{e_1} \cdots B_{e_{s-1}} \\ \mid Q_{o_{s+1}} \cdots Q_{o_k} \mid Q_{e_k} \cdots Q_{e_s} \mid A_{e_{s-1}} \cdots A_{e_1},$$

with pivots $(1, 2s-1)$, and for $1 \leq s \leq k-1$ define

$$L_{\text{even}}(k, s) = A_{o_1} \cdots A_{o_{s+1}} \mid B_{o_{s+1}} \mid Q_{o_{s+2}} \cdots Q_{o_k} \\ \mid Q_{e_k} \cdots Q_{e_{s+1}} \mid B_{o_s} \cdots B_{o_1} \mid B_{e_1} \cdots B_{e_s} \mid A_{e_s} \cdots A_{e_1},$$

with pivots $(1, 2s)$.

For $1 \leq s \leq k$ define

$$R_{\text{odd}}(k, s) = A_{o_k} \cdots A_{o_s} \mid B_{o_s} \mid Q_{o_{s-1}} \cdots Q_{o_1} \mid Q_{e_1} \cdots Q_{e_{s-1}} \\ \mid B_{o_{s+1}} \cdots B_{o_k} \mid B_{e_k} \cdots B_{e_s} \mid A_{e_s} \cdots A_{e_k},$$

with pivots $(m, 2s - 1)$, and for $1 \leq s \leq k - 1$ define

$$R_{\text{even}}(k, s) = A_{o_k} \cdots A_{o_{s+1}} \mid B_{o_{s+1}} \cdots B_{o_k} \mid B_{e_k} \cdots B_{e_{s+1}} \\ \mid Q_{o_s} \cdots Q_{o_1} \mid Q_{e_1} \cdots Q_{e_s} \mid A_{e_{s+1}} \cdots A_{e_k},$$

with pivots $(m, 2s)$.

The index ranges partition the four label classes, so every displayed word is a strict total order of all $4k$ labels. Deleting all B labels or all A labels gives exactly the claimed row-normal-form words. The true formula overlaps are $(1, 1)$ and (m, m) , where the diagonal and boundary words are literally identical. The pairs $(1, m)$ and $(m, 1)$ are ordinary boundary members of $L_{\text{odd}}(k, k)$ and $R_{\text{odd}}(k, 1)$. All formulas remain literal at $k = 2$ under the empty-range convention.

D.4 Horizontal stack validity of the constructions

For $MMMMVV$, the horizontal actions are

$$A_{o_r} = \text{push}(o_r), \quad B_{o_r} = \text{pop}(o_r), \quad B_{e_r} = \text{push}(e_r), \quad A_{e_r} = \text{pop}(e_r).$$

Each Q_c is therefore an immediate push/pop pair above the pre-existing stack. The five construction families can be checked literally.

- **Diagonal** $D(k, p)$. Every Q_{c_j} pushes and immediately pops c_j , so the stack is empty before and after every block.
- **Left odd** $L_{\text{odd}}(k, s)$. The initial $A_{o_1}, \dots, A_{o_s}$ pushes the odd core $(o_1, \dots, o_s)$, and $B_{o_s}, \dots, B_{o_1}$ pops it. Then $B_{e_1}, \dots, B_{e_{s-1}}$ pushes the even core $(e_1, \dots, e_{s-1})$. All later Q blocks act above that core, and $A_{e_{s-1}}, \dots, A_{e_1}$ pops it.
- **Left even** $L_{\text{even}}(k, s)$. The first odd run pushes $(o_1, \dots, o_{s+1})$; $B_{o_{s+1}}$ pops its top, leaving $(o_1, \dots, o_s)$. The intervening odd and even Q blocks act above that core. Then $B_{o_s}, \dots, B_{o_1}$ empties the odd core, $B_{e_1}, \dots, B_{e_s}$ pushes a fresh even core, and $A_{e_s}, \dots, A_{e_1}$ pops it.
- **Right odd** $R_{\text{odd}}(k, s)$. The first run pushes $(o_k, o_{k-1}, \dots, o_s)$; B_{o_s} removes the top. The Q blocks act above the residual odd core $(o_k, \dots, o_{s+1})$, which is then popped by $B_{o_{s+1}}, \dots, B_{o_k}$. Finally $B_{e_k}, \dots, B_{e_s}$ pushes the even core $(e_k, \dots, e_s)$ and $A_{e_s}, \dots, A_{e_k}$ pops it.
- **Right even** $R_{\text{even}}(k, s)$. The initial odd core $(o_k, \dots, o_{s+1})$ is pushed and then popped completely. The run $B_{e_k}, \dots, B_{e_{s+1}}$ pushes an even core; all Q blocks act above it; the final run $A_{e_{s+1}}, \dots, A_{e_k}$ pops it.

All empty endpoint ranges are literal, so the same verification covers $s = 1$, $s = k$, and $s = k - 1$ where those parameters are legal. In every family there is no underflow, every closer matches the current top, and the final stack is empty. Consequently every horizontal directed relation holds and every two horizontal intervals are disjoint or nested, never strictly alternating.

D.5 Remaining U/L full-validity obligations

Proposition 6.1 already handles all directed vertical relations and same-row equal-target pairs. H-stack validity handles the horizontal system. Therefore only U_i/L_j with equal seam parity remain.

For the diagonal family, each column is a contiguous block Q_c in the row-normal-form column order W_p . If $i \neq j$ have equal parity, then the adjacent-column pairs $\{i, i+1\}$ and $\{j, j+1\}$ are disjoint. Strict alternation of the actual U/L endpoints would force alternation of those column pairs in W_p , contradicting row-local validity from Proposition 6.1. For $i = j$, the two adjacent column blocks occur with opposite internal orientations, and direct inspection shows one vertical interval nested in the other.

For a boundary family fix the upper A chain. Let $\rho(c)$ be the position of A_c , and put

$$\sigma(c) = \#\{A_d : A_d \prec B_c\}.$$

Lemma D.1 (Slots and cross-row alternation). *Suppose an upper seam has endpoint ranks $\alpha < \beta$ in the upper A chain and a lower seam has endpoint slots $x \leq y$. The two seams strictly alternate if and only if exactly one of x, y belongs to the integer interval $[\alpha, \beta - 1]$. In particular, alternation is impossible if $x = y$, or if $x + y = \alpha + \beta - 1$.*

Proof. A lower face lies strictly between the upper endpoints exactly when at least α , but fewer than β , upper faces precede it, i.e. its slot belongs to $[\alpha, \beta - 1]$. Two intervals with four distinct endpoints alternate exactly when precisely one endpoint of the second interval lies inside the first. This gives the criterion and the same-slot assertion. If $x + y = \alpha + \beta - 1$ and $x \geq \alpha$, then $y \leq \beta - 1$, so both slots lie inside. If $x < \alpha$, integrality gives $x \leq \alpha - 1$ and hence $y \geq \beta$, so the lower endpoints lie on the two exterior sides. Neither case gives exactly one interior endpoint. $\square$

For upper chain W_1 ,

$$\rho_1(o_r) = r, \quad \rho_1(e_r) = 2k - r + 1,$$

so

$$S_r^o = [r, 2k - r] \quad (1 \leq r \leq k), \quad S_r^e = [r + 1, 2k - r] \quad (1 \leq r \leq k - 1).$$

For upper chain W_m ,

$$\rho_m(o_r) = k - r + 1, \quad \rho_m(e_r) = k + r,$$

so

$$T_r^o = [k - r + 1, k + r - 1] \quad (1 \leq r \leq k), \quad T_r^e = [k - r, k + r - 1] \quad (1 \leq r \leq k - 1).$$

The sums of the two endpoints in these four shell families are, respectively, $2k$, $2k + 1$, $2k$, and $2k - 1$.

The required σ values are as follows; the three columns correspond to $t < s$, $t = s$, and $t > s$.

construction	column	$t < s$	$t = s$	$t > s$
$L_{\text{odd}}(k, s)$	$\sigma(o_t)$	s	s	t
	$\sigma(e_t)$	s	$2k - s$	$2k - t$
$L_{\text{even}}(k, s)$	$\sigma(o_t)$	$2k - s$	$2k - s$	t
	$\sigma(e_t)$	$2k - s$	$2k - s$	$2k - t$
$R_{\text{odd}}(k, s)$	$\sigma(o_t)$	$k - t + 1$	$k - s + 1$	$k + s - 1$
	$\sigma(e_t)$	$k + t - 1$	$k + s - 1$	$k + s - 1$
$R_{\text{even}}(k, s)$	$\sigma(o_t)$	$k - t + 1$	$k - s + 1$	$k - s$
	$\sigma(e_t)$	$k + t - 1$	$k + s - 1$	$k - s$

construction	column	$t < s$	$t = s$	$t > s$
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These values are obtained directly by counting preceding A labels in the displayed construction words. They give the following complete lower-seam classification.

- In L_{odd} , odd seam $2t - 1$ is a point for $t < s$, is S_s^o for $t = s$, and is S_t^o for $t > s$; even seam $2t$ is a point for $t < s$, is S_s^e for $t = s$, and is S_t^e for $t > s$.
- In L_{even} , odd seam $2t - 1$ is a point for $t \leq s$ and is S_t^o for $t > s$; even seam $2t$ is a point for $t < s$, is S_s^e for $t = s$, and is S_t^e for $t > s$.
- In R_{odd} , odd seam $2t - 1$ is T_t^o for $t \leq s$ and a point for $t > s$; even seam $2t$ is T_t^e for $t < s$ and a point for $t \geq s$.
- In R_{even} , odd seam $2t - 1$ is T_t^o for $t \leq s$ and a point for $t > s$; even seam $2t$ is T_t^e for $t \leq s$ and a point for $t > s$.

Thus every non-point lower seam is exactly a member of the corresponding upper shell family. For an arbitrary upper seam of the same parity, its shell and the two lower endpoint slots belong to the same one of the four families above and therefore have the same endpoint sum. If the upper shell is $[\alpha, \beta - 1]$, the lower slots satisfy $x + y = \alpha + \beta - 1$, so Lemma D.1 excludes strict alternation. Point seams are excluded by the same lemma. This proves all remaining U_i/L_j obligations for arbitrary separation of the seam indices. The argument uses the constant endpoint sum; numerical containment of slot intervals by itself is not used.

D.6 H-local legality is not H-completability

For fixed row chains, represent a partial H merge by consumed row-prefix lengths and the stack S of open columns. A next row head is H-locally legal if it obeys one push/pop step. It is H-completable only if the child state admits a full continuation ending with every face consumed and the stack empty.

These notions differ. At $k = 2$ with $(a, b) = (1, 1)$, after prefix A_1 both A_3 and B_1 are locally legal, but only B_1 admits completion. Therefore the uniqueness proof never uses the false rule “every locally legal next event is globally forced.”

Lemma D.2 (Diagonal fixed-row uniqueness). *For the canonical assignment $MMMMVV$ and every $1 \leq p \leq m$, the two row orders with pivots (p, p) have exactly one complete H-valid merge, namely $D(k, p)$.*

Proof. Write their common column order as $W_p = (c_1, \dots, c_n)$. Assume inductively that the first $r - 1$ columns have been processed as $Q_{c_1} \cdots Q_{c_{r-1}}$. The horizontal stack is empty and both row heads have column c_r . The opener must be emitted first: it is A_{c_r} for odd c_r , and B_{c_r} for even c_r .

The other row is now waiting at its matching closer. Suppose the row that emitted the opener advances instead to its next column d . If that next event is a closer, its matching opener has not appeared, since it occurs after c_r in the blocked opposite row. If it is an opener, its label d is pushed above c_r . In any complete LIFO trace, d would then have to close before c_r , whereas the opposite row requires the closer of c_r before the closer of d . Both alternatives are impossible. If this row is exhausted, there is no alternative event. Hence the matching closer follows immediately, producing Q_{c_r} , emptying the stack, and restoring the induction hypothesis. The resulting word is $D(k, p)$, which is H-valid by its immediate push-pop blocks. The proof includes $p = 1, m$ as well as interior pivots. $\square$

D.7 Projected determinism and first disagreement

In any complete H-valid merge, deleting all events of one column parity leaves a legal stack trace. For the odd projection, the upper row fixes the opener stream and the lower row fixes the closer stream. For the even projection these roles are exchanged. A fixed opener stream and a fixed closer stream have at most one complete legal stack trace: if the next requested closer has not opened, the opener stream is forced through that label; once it is the top label, closing it is necessary, since another push would bury the fixed next closer. A requested closer that is already buried cannot be completed. Thus both parity projections are uniquely determined whenever they exist.

For a boundary pivot pair, let T be the explicit H-valid construction from D.3–D.4. Suppose a different complete H-valid merge T' exists, and consider their first divergence. Their common prefix is a prefix of the known word T . An upper-odd/lower-odd divergence contradicts uniqueness of the odd projection, and an upper-even/lower-even divergence contradicts uniqueness of the even projection. An upper-even/lower-odd divergence would require two distinct closers to be simultaneously legal at one stack top. The only remaining possibility is an upper-odd opener versus a lower-even opener.

D.8 Opener windows on the explicit constructions

The following table lists all prefixes of the four explicit boundary constructions at which both row heads are openers. In the stack column, entries are written from bottom to top.

T	range	stack	row heads
$L_{\text{odd}}(k, s)$	$2 \leq s \leq k-1, 1 \leq t \leq s-1$	$(e_1, \dots, e_{t-1})$	$(A_{o_{s+1}}, B_{e_t})$
$L_{\text{even}}(k, s)$	none	—	—
$R_{\text{odd}}(k, s)$	none	—	—
$R_{\text{even}}(k, s)$	$1 \leq s \leq k-1, s+1 \leq t \leq k$	$(e_k, e_{k-1}, \dots, e_{t+1})$	(A_{o_s}, B_{e_t})

In every listed window, T chooses the lower-even opener. For L_{odd} , the initial odd core is opened and closed before the lower even core $e_1, \dots, e_{s-1}$ is opened. During this last phase the next upper face is $A_{o_{s+1}}$ exactly when $s < k$. The range is empty for $s = 1$, and for $s = k$ the upper head is already an even closer. For R_{even} , the initial odd core is likewise emptied; the lower even core is then opened from e_k down to e_{s+1} while the upper head remains A_{o_s} , giving the second displayed range. For L_{even} and R_{odd} , all upper odd openers have appeared before the first lower even opener is reached. Immediate Q_c blocks and the final closing phases create no further opener/opener windows. This exhausts prefixes of the explicit constructions, not arbitrary H-locally legal dead-end prefixes. That is exactly the scope required by the first-divergence argument.

D.9 Wrong-switch barriers and completion of uniqueness

At a left-odd window, suppose T' chooses $A_{o_{s+1}}$ instead of B_{e_t} . Before the lower row can reach $B_{o_{s+1}}$, it must open $e_t, \dots, e_{s-1}$ above the newly opened odd label. Consequently LIFO requires $A_{e_{s-1}} \prec B_{o_{s+1}}$. The fixed row orders and horizontal precedence then give the impossible cycle

$$B_{o_{s+1}} \prec_B B_{e_k} \prec_H A_{e_k} \prec_A A_{e_{s-1}} \prec_{\text{LIFO}} B_{o_{s+1}}.$$

Here $\prec_A, \prec_B$ are the prescribed upper and lower row orders; $\prec_H$ is horizontal opener-before-closer precedence.

At a right-even window, choosing A_{o_s} forces the later lower opener e_{s+1} above it before B_{o_s} can appear. LIFO requires $A_{e_{s+1}} \prec B_{o_s}$, while the other constraints give

$$B_{o_s} \prec_B B_{e_1} \prec_H A_{e_1} \prec_A A_{e_{s+1}} \prec_{\text{LIFO}} B_{o_s}.$$

The index ranges in the window table ensure that all events used in these cycles exist. Thus a first divergence from T is impossible. Each boundary pair therefore has at most one complete H-valid merge. Lemma D.2 covers every diagonal pair. Together these cases cover all pairs $a = 1$, $a = m$, or $a = b$. The constructions give existence and D.5 gives full validity, completing the multiplicity-one claim without using a count formula.

D.10 Global-state-to-pivot bridge and multiplicity

Every globally valid $MMMMVV$ state obeys all row-local constraints. Proposition 6.1 therefore assigns its upper and lower restrictions unique pivots (a, b) . This is a coverage/parameterization statement only; it does not itself assert compatibility or existence.

Necessity shows the recovered pair is allowed. Sections D.3–D.5 give one fully valid state for each allowed pair. Lemma D.2 gives fixed-pair uniqueness on the diagonal, while Sections D.7–D.9 give fixed-pair uniqueness for the boundary families. These cases exhaust $a = 1$, $a = m$, or $a = b$. Hence the global-state-to-pair map is a bijection and each compatible pair has multiplicity one. Only after this bridge may the compatible-pair cardinality be identified with the state count.

E Reconstruction and block-compatibility closure for $MVMMMM$

This appendix keeps the two logical ingredients of Theorem 7.2 separate: fixed-row reconstruction/full validity, and the independent block criterion.

E.1 Fixed-row semantics

For $MVMMMM$, every horizontal relation is

$$A_c \prec B_c.$$

The upper row is $R^+(k, a)$ and the lower row is $R^-(k, b)$. Hence the upper column sequence is the H-opener stream, while the lower column sequence is the H-closer request stream.

E.2 Fixed-row reconstruction: H/H noncrossing is LIFO

With each opener before its matching closer, two horizontal intervals cross exactly when an earlier-opened interval closes while a later-opened distinct interval is still open. Pairwise H/H Butterfly noncrossing is therefore equivalent to the usual LIFO stack rule.

Fix the two row chains and let c be the next requested lower label. If the matching upper opener has not yet been emitted, no lower label may bypass c , so the upper chain is forced until that opener appears. If c is the stack top, its closer is forced immediately; pushing another opener would block the fixed next closer. If c is already buried in the stack, no completion exists. Induction over the lower events proves that an H-compatible merge is unique when it exists, and that the forced reconstruction succeeds exactly when H compatibility holds.

E.3 Automatic remainder after horizontal reconstruction

Proposition 6.1 supplies all upper and lower vertical directed constraints and all same-row same-target vertical noncrossing. The H stack supplies every horizontal precedence and every H/H

noncrossing condition. Since the horizontal target S is distinct from the vertical targets E, W , no mixed horizontal/vertical twin family occurs. The only remaining class is U_i/L_j with equal seam parity.

For seam i , orient the adjacent horizontal nesting as

$$A_{\alpha_i} \prec A_{\beta_i} \prec B_{\beta_i} \prec B_{\alpha_i},$$

where $(\alpha_i, \beta_i) = (i, i+1)$ for odd i and $(i+1, i)$ for even i . Thus U_i is the opener-side interval and L_i the closer-side interval of the same nested H pair.

For distinct equal-parity seams i, j , the two adjacent column pairs are disjoint. Consider two nested H pairs

$$o_x < o_y < c_y < c_x, \quad o_p < o_q < c_q < c_p,$$

with opener-side intervals $U_e = [o_x, o_y]$, $U_f = [o_p, o_q]$ and closer-side intervals $L_e = [c_y, c_x]$, $L_f = [c_q, c_p]$. Suppose U_e crosses L_f .

If

$$c_q < o_x < c_p < o_y,$$

then

$$o_p < o_x < c_p < c_x,$$

so the horizontal intervals $[o_p, c_p]$ and $[o_x, c_x]$ cross, contradicting horizontal laminarity. Hence the only possible alternating order is

$$o_x < c_q < o_y < c_p.$$

If $o_p < o_x$, horizontal laminarity of $[o_q, c_q]$ with $[o_x, c_x]$ forces $o_x < o_q$, giving

$$o_p < o_x < o_q < o_y,$$

so U_e crosses U_f . If instead $o_x < o_p$, laminarity of $[o_p, c_p]$ with $[o_y, c_y]$ forces $c_y < c_p$, while laminarity of $[o_x, c_x]$ with $[o_p, c_p]$ forces $c_p < c_x$. Together with $c_q < o_y < c_y$ this yields

$$c_q < c_y < c_p < c_x,$$

so L_e crosses L_f . Therefore

$$U_i \text{ crosses } L_j \implies (U_i \text{ crosses } U_j) \text{ or } (L_i \text{ crosses } L_j).$$

Equal parity gives the same vertical target class, and Proposition 6.1 forbids both same-row alternatives. For $i = j$, the upper interval lies in the opener side of the nested adjacent H pair and the lower interval in the closer side, so they are disjoint. Thus every H-compatible merge is globally valid:

$$\text{FullCompat}(k; a, b) \iff \text{HCompat}(k; a, b).$$

E.4 Endpoint normal forms for the block criterion

Let W_p denote the $+$ row-normal-form column word for pivot p . Because the lower row is the reverse normal form, its closer request stream is

$$\overleftarrow{W}_p := \text{rev}(W_p).$$

The singleton-block word is

$$W_1 = (1, 3, 5, \dots, 2k-1, 2k, 2k-2, \dots, 2).$$

For $1 \leq r \leq k-1$, name the two pivots in block r by

$$p_r^{(0)} = 2r, \quad p_r^{(1)} = 2r+1.$$

Set $c_r = 2r+1$, $d_r = 2r+2$ and

$$L_r = (2r-1, 2r-3, \dots, 1, 2, 4, \dots, 2r),$$

$$R_r = (2r+3, 2r+5, \dots, 2k-1, 2k, 2k-2, \dots, 2r+4).$$

Then

$$W_{p_r^{(0)}} = c_r R_r d_r L_r, \quad W_{p_r^{(1)}} = c_r L_r R_r d_r,$$

and

$$\overleftarrow{W}_{p_r^{(0)}} = \text{rev}(L_r) d_r \text{rev}(R_r) c_r,$$

$$\overleftarrow{W}_{p_r^{(1)}} = d_r \text{rev}(R_r) \text{rev}(L_r) c_r.$$

At $r = k-1$, R_r is empty and the formulas remain literal. The block coordinate records the common junction index:

$$B_R(1) = 0, \quad B_R(p_r^{(0)}) = B_R(p_r^{(1)}) = r.$$

E.5 Necessity for the block criterion: unequal blocks

For three distinct horizontal labels x, y, z , suppose the opener order contains

$$y \prec_W z \prec_W x$$

while the closer stream requires

$$x \prec_{\leftarrow W} y \prec_{\leftarrow W} z.$$

No LIFO schedule can realize both. When x closes, both y and z have already opened; after x is removed, z lies above y , yet the prescribed closer stream asks for y before z . Additional pushes cannot repair the order and z is not permitted to close first.

Every unequal ordered block pair contains such a witness. Since $\overleftarrow{W}_b = \text{rev}(W_b)$, it is enough to verify

$$y \prec_{W_a} z \prec_{W_a} x, \quad z \prec_{W_b} y \prec_{W_b} x.$$

The following table covers all ordered unequal-block cases directly.

ordered case	range	(x, y, z)	order in W_a	order in W_b
$1/p_s^{(0)}$	$1 \leq s \leq k-1$	$(2, 1, 2s+1)$	$1 < 2s+1 < 2$	$2s+1 < 1 < 2$
$1/p_s^{(1)}$	$1 \leq s \leq k-1$	$(2, 1, 2s+1)$	$1 < 2s+1 < 2$	$2s+1 < 1 < 2$
$p_r^{(0)}/1$	$1 \leq r \leq k-1$	$(2, 2r+1, 1)$	$2r+1 < 1 < 2$	$1 < 2r+1 < 2$
$p_r^{(1)}/1$	$1 \leq r \leq k-1$	$(2, 2r+1, 1)$	$2r+1 < 1 < 2$	$1 < 2r+1 < 2$
$p_r^{(0)}/p_s^{(0)}$ or $p_r^{(0)}/p_s^{(1)}$	$r < s$	$(1, 2r+1, 2r+3)$	$2r+1 < 2r+3 < 1$	$2r+3 < 2r+1 < 1$
$p_r^{(1)}/p_s^{(0)}$ or $p_r^{(1)}/p_s^{(1)}$	$r < s$	$(2s, 1, 2r+3)$	$1 < 2r+3 < 2s$	$2r+3 < 1 < 2s$

ordered case	range	(x, y, z)	order in W_a	order in W_b
$p_r^{(0)}/p_s^{(0)}$ or $p_r^{(1)}/p_s^{(0)}$	$r > s$	$(1, 2r + 1, 2r - 1)$	$2r + 1 < 2r - 1 < 1$	$2r - 1 < 2r + 1 < 1$
$p_r^{(0)}/p_s^{(1)}$ or $p_r^{(1)}/p_s^{(1)}$	$r > s$	$(2r, 2r + 1, 1)$	$2r + 1 < 1 < 2r$	$1 < 2r + 1 < 2r$

All witnesses lie in range. If $r < s$, then $r \leq k - 2$ and $2r + 3 \leq 2k - 1$; if $r > s$, then $r \geq 2$ and $2r - 1 \geq 3$; singleton/non-singleton cases require only $k \geq 2$. Hence every unequal-block pair fails H compatibility.

E.6 Sufficiency for the block criterion: same-block schedules

For the singleton pair $(1, 1)$, push all openers in W_1 and pop them in reverse order.

Within a nonzero block r , the diagonal pairs

$$(p_r^{(0)}, p_r^{(0)}), \quad (p_r^{(1)}, p_r^{(1)})$$

again use push-all/pop-all. For $(p_r^{(0)}, p_r^{(1)})$, use

$$\text{push } c_r, R_r, d_r; \quad \text{pop } d_r, \text{rev}(R_r); \quad \text{push } L_r; \quad \text{pop rev}(L_r); \quad \text{pop } c_r.$$

For $(p_r^{(1)}, p_r^{(0)})$, use

$$\text{push } c_r, L_r; \quad \text{pop rev}(L_r); \quad \text{push } R_r, d_r; \quad \text{pop } d_r, \text{rev}(R_r); \quad \text{pop } c_r.$$

Every requested closer is the stack top, every label is used once, and the stack ends empty. Thus equal block coordinate implies H compatibility.

Combining E.5 and E.6 gives

$$\text{HCompat}(k; a, b) \iff B_R(a) = B_R(b).$$

Together with E.3 this proves Theorem 7.2. No finite compatibility graph or count formula is used as a premise.

F Detailed Boundary Proof and the Eight Physical Singleton States

This appendix uses only the $n = 2$ physical map.

F.1 Physical projection and crease census

The crease-incidence cycle is

$$A_1 - H_1 - B_1 - L_1 - B_2 - H_2 - A_2 - U_1 - A_1.$$

The four physical labels are h_0, h_1, u_0, l_0 . Hence the assigned map depends only on $\pi_1(P) = (h_0, h_1, u_0, l_0)$, and each physical assignment has four formal representatives obtained by arbitrary choices of the two absent formal bits.

The complete global test consists of the four primitive M/V precedences and exactly two Butterfly pairs: (H_1, H_2) and (U_1, L_1) . These are the two pairs of opposite edges of the cycle.

F.2 Negative product: Hamiltonian uniqueness and validity

Traverse the cycle as $A_1 \rightarrow B_1 \rightarrow B_2 \rightarrow A_2 \rightarrow A_1$ and assign sign $+1$ to a primitive edge pointing with this traversal and -1 to an edge pointing against it. Direct checkerboard evaluation gives traversal signs

$$h_0, \quad -l_0, \quad h_1, \quad -u_0,$$

whose product is $h_0 h_1 u_0 l_0 = \delta_0$.

If $\delta_0 = -1$, exactly one or three edges oppose the traversal. In either orientation, the three majority edges form a directed Hamiltonian path $v_0 \prec v_1 \prec v_2 \prec v_3$ and the remaining edge is the shortcut $v_0 \prec v_3$. Thus there is exactly one linear extension. The two opposite edge pairs have endpoint intervals $[0, 1]$ with $[2, 3]$ and $[1, 2]$ with $[0, 3]$, respectively; one pair is disjoint and the other nested. Hence the unique order passes both Butterfly tests.

F.3 Positive product: direct global obstruction

If $\delta_0 = +1$, the number of traversal reversals is 0, 2, or 4. With 0 or 4 reversals the primitive precedences form a directed 4-cycle. With exactly two reversals, the orientation is acyclic but every linear extension makes one opposite-edge pair alternate.

If the two reversed edges are adjacent, the poset has one source, one sink, and two incomparable middle vertices. The two possible middle orders force intervals $[0, 2]$ and $[1, 3]$ for one of the opposite-edge pairs. If the reversed edges are opposite, the poset has two sources followed by two sinks; whichever internal source/sink order is chosen, one opposite-edge pair again has intervals $[0, 2]$ and $[1, 3]$. Hence no globally valid state exists.

F.4 Exact physical state table

physical projection (h_0, h_1, u_0, l_0)	exact state set
<i>MMMV</i>	$\{(A_1, B_1, B_2, A_2)\}$
<i>MMVM</i>	$\{(B_2, A_2, A_1, B_1)\}$
<i>MVMM</i>	$\{(A_1, A_2, B_2, B_1)\}$
<i>MVVV</i>	$\{(A_2, A_1, B_1, B_2)\}$
<i>VMMM</i>	$\{(B_2, B_1, A_1, A_2)\}$
<i>VMVV</i>	$\{(B_1, B_2, A_2, A_1)\}$
<i>VVMV</i>	$\{(B_1, A_1, A_2, B_2)\}$
<i>VVVM</i>	$\{(A_2, B_2, B_1, A_1)\}$

The other eight physical projections have empty exact state set. The table is a literal presentation of the analytic Hamiltonian rule, not a computational premise.